

Dr.Docker :A Large-Scale Security measurement of Docker Image Ecosystem.

Team Details

- | | | |
|----|------------------------|--------------|
| 1. | GANDI VARSHINI | (22EG105B17) |
| 2. | AENUGU MANISH REDDY | (22EG105B12) |
| 3. | CHIRVA SAI KOUSHIK | (22EG105B10) |
| 4. | KOLLA WAJJALA SREDDUTT | (22EG105B31) |

Project Supervisor:

DR.K.MADHURI
Associate Professor

Introduction

What it is :

Docker images are widely shared and reused to build applications across cloud and DevOps environments. However, many images contain security issues such as outdated software, exposed secrets, and unsafe configurations. Most earlier work studied only small sets of images or focused on one problem at a time, so we still do not clearly understand the overall security condition of Docker images.

Our project, **Dr.Docker**, performs **large-scale measurement and analysis** of Docker images to study their security status and highlight areas where the ecosystem can be improved.

What is needed:

There is a need for a system that can **analyze Docker images at large scale** and show the real security condition of the ecosystem. Existing studies usually focus on small datasets or single problems, so they do not give a complete picture. A combined analysis approach is needed to understand how different security issues appear and spread across images.

What are applications:

This system helps developers and organizations choose safer Docker images and use them more responsibly in cloud and DevOps environments.

It also supports researchers in studying security patterns across the Docker ecosystem and encourages better image-building practices and isolation techniques.

Literature Survey

Ref No	Author Name	Journal / Conference	Findings	Limitations + Our Approach
[1]	Markus Dahlmanns et al.	ACM Asia CCS 2023	Identified large-scale secret leakage such as API keys and credentials in container images published in public registries.	Focused only on secret leakage. Our work analyzes multiple security issues together.
[2]	Zeng W. et al.	ACM CNML 2023	Studied container security management and showed that misconfigurations and poor isolation increase security risks.	Focused on runtime security. Our work analyzes Docker images before deployment.
[3]	Thomas Durieux	IEEE/ACM ICSE 2024	Analyzed Docker image smells and outdated dependencies that increase image size and reduce maintainability.	Focused on quality issues, not direct security risks. Our work focuses on security analysis of images.
[4]	Hamid Mohayjeji et al.	IEEE ICSME 2025	Studied security vulnerabilities across Docker image tags and showed how dependencies influence vulnerability propagation.	Limited to cross-tag analysis. Our work studies ecosystem-level dependency impact.
[5]	Nousias A. et al.	IEEE 2025	Proposed techniques to detect malicious activities and malware inside container environments.	Focused mainly on malware detection. Our work covers vulnerabilities, leaks, and misconfigurations.

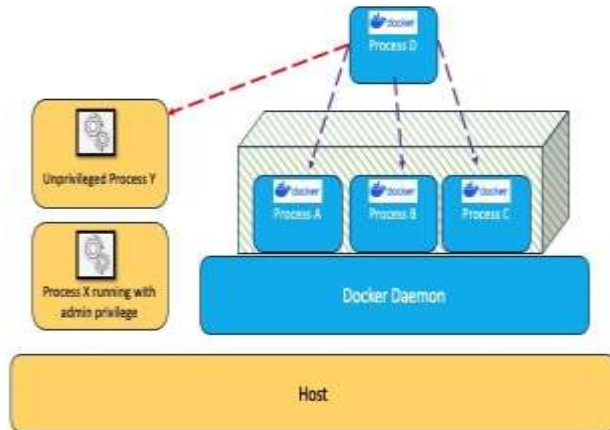
Problem Statement

- **Docker images are widely reused** in cloud and DevOps environments, often without proper security checking. Many images contain **hidden security issues** such as vulnerabilities, leaked secrets, outdated software, and unsafe configurations.
Since new images are usually built using existing images, security problems in one image can spread to many other images.
- Most existing research studies examine Docker images individually or focus on only one security issue. As a result, it is difficult to understand the overall security condition of the Docker image ecosystem and how security risks spread through **image dependencies**.
Therefore, a large-scale and systematic analysis of Docker images is needed to better assess ecosystem-level security risks.

OBJECTIVES

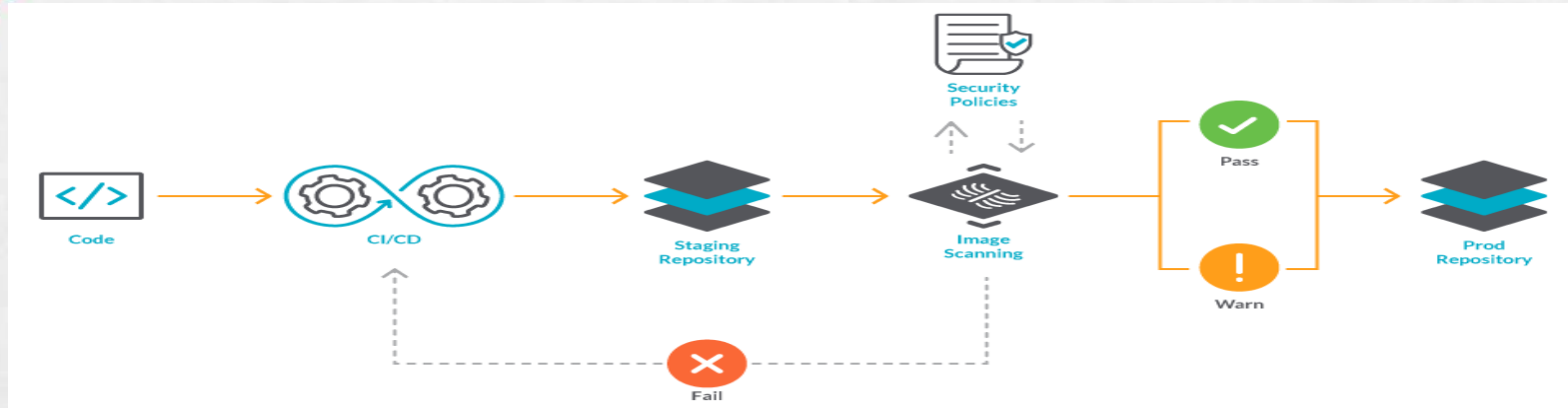
- ❖ **To collect Docker images from public repositories for analysis:** focusing on collecting Docker images from public repositories and building a dataset for analysis. The images will be downloaded automatically so that real-world images are included instead of only sample files. This dataset forms the base for studying security across many Docker images.
- ❖ **To scan each Docker image and identify security issues:** the collected Docker images are scanned and examined to identify possible security issues. The analysis process detects vulnerabilities, outdated components, and misconfigurations in each image. This helps in understanding how secure or insecure an image is before it is used.
- ❖ **To calculate security metrics for every image:** After scanning, security information from each image is processed and converted into measurable values. We record the number of vulnerabilities, classify them by severity, and generate a simple risk score. These values make it easier to compare different images and understand their overall security level.
- ❖ **To generate simple reports that help users avoid high-risk images:** The analyzed results are summarized and presented in a clear and user-friendly way. The system displays tables or graphs showing which images are safer and which are risky.

Proposed Methodology



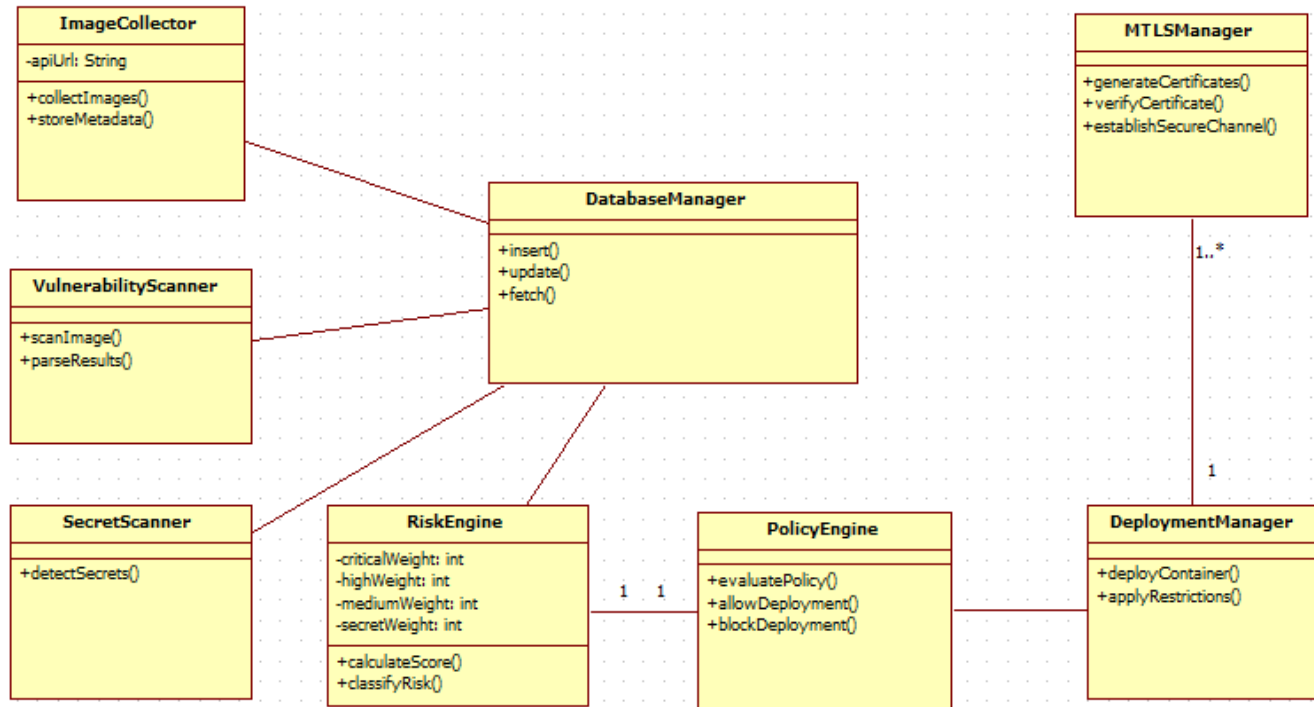
- ❖ The process starts with **selection of Docker images** from public container repositories.
- ❖ The selected images are **automatically downloaded** and stored to create a structured dataset.
- ❖ An **automated analysis framework** processes each Docker image without manual intervention.
- ❖ Each image is **scanned to detect security issues**, such as software vulnerabilities, outdated components, and configuration risks.

Proposed Methodology

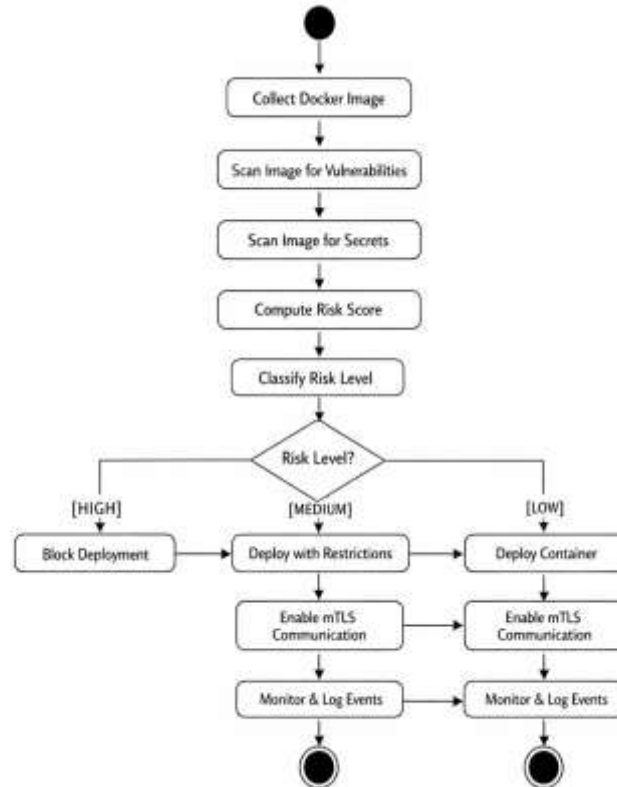


- ❖ The detected issues are **converted into security metrics**, including vulnerability count, severity classification, and an overall risk score.
- ❖ All results are **aggregated and analyzed collectively** to understand security trends across images.
- ❖ Finally, the outcomes are **presented as simple reports** using tables or graphs to clearly identify high-risk and low-risk Docker images.

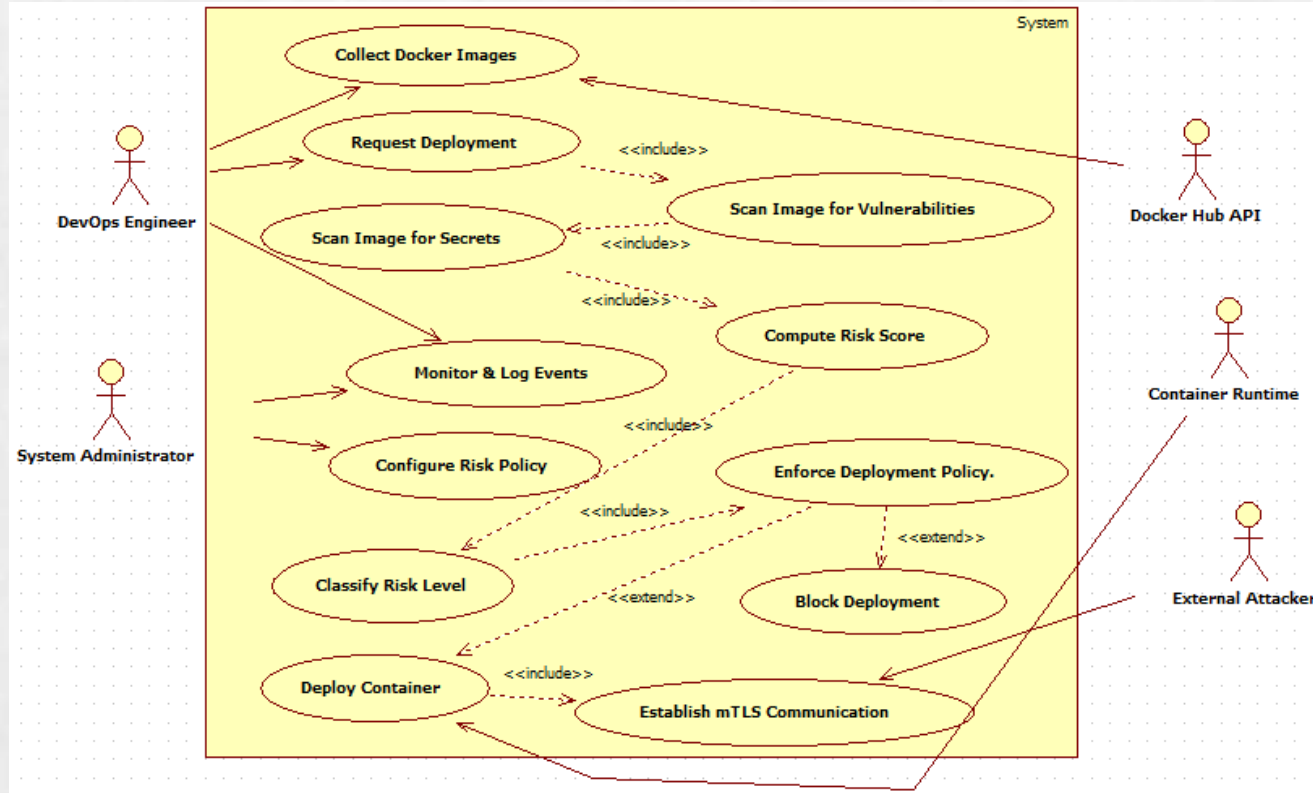
Class Diagram



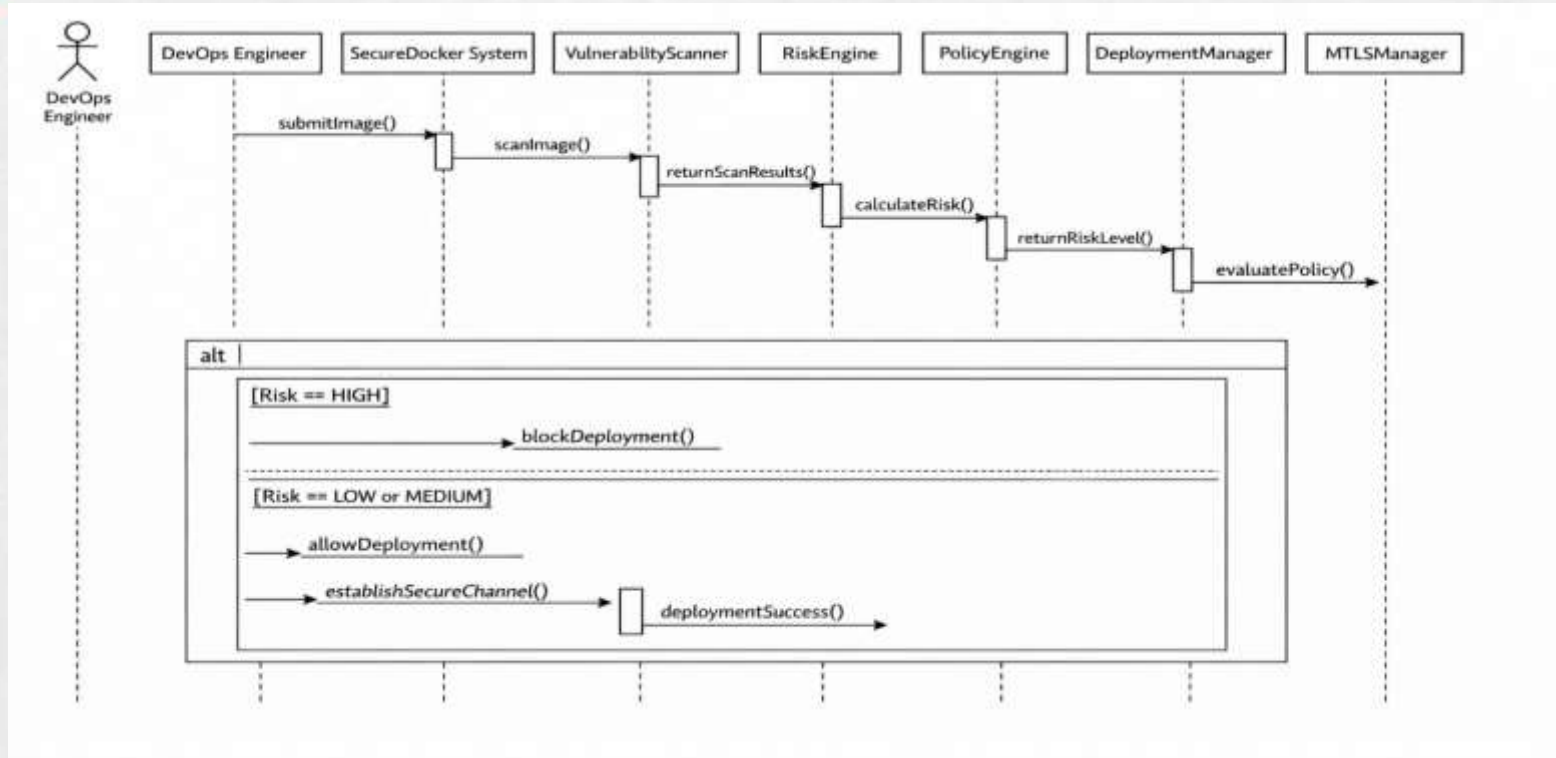
Activity Diagram



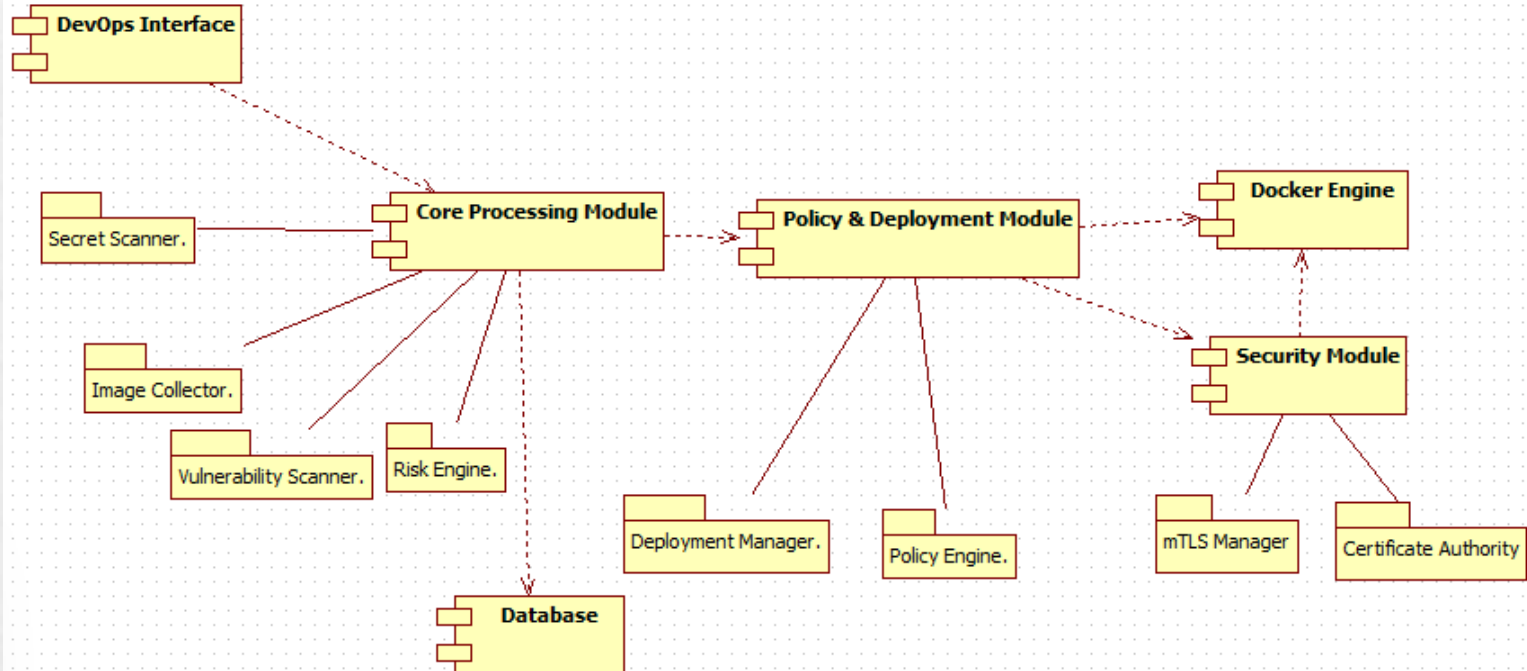
Use Case Diagram



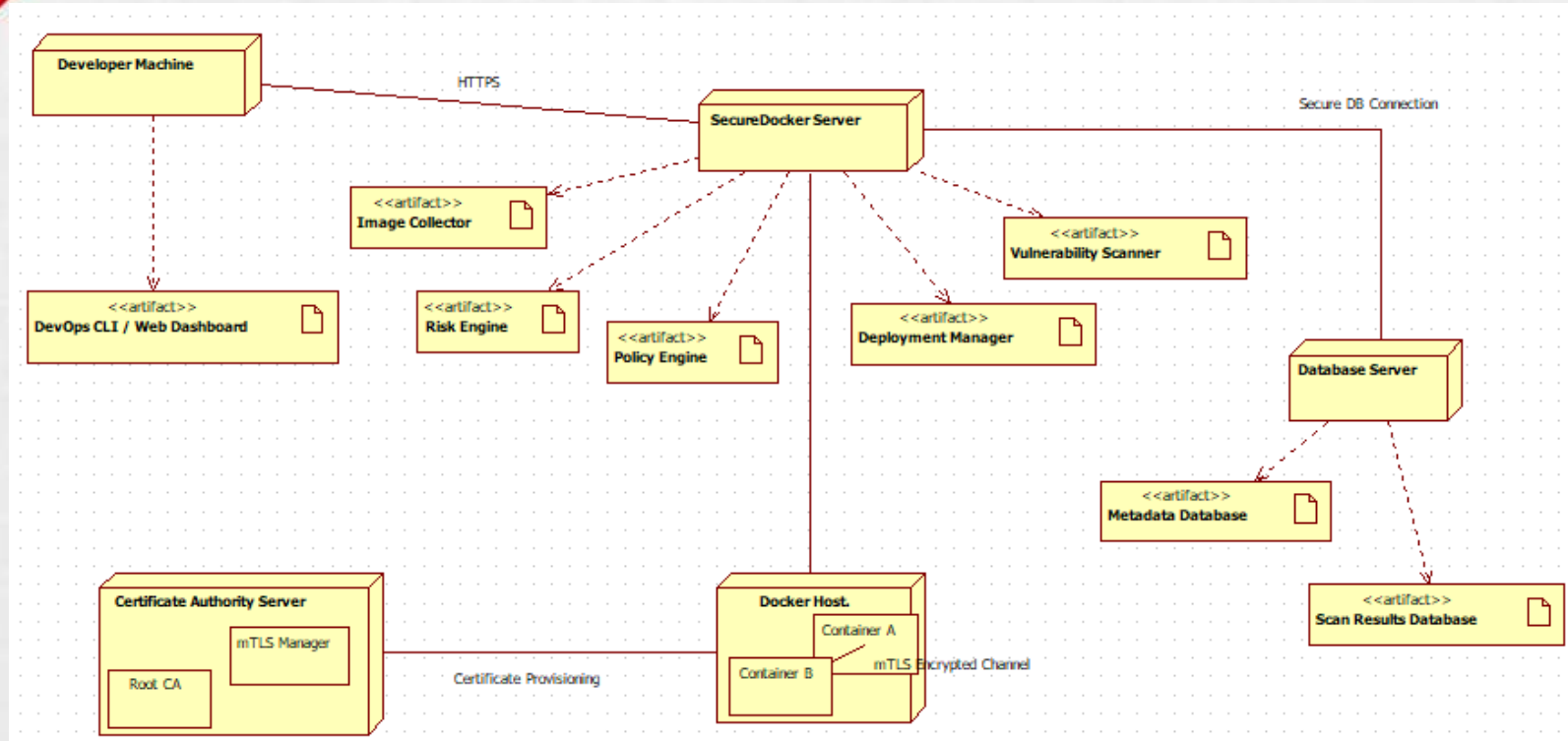
Sequence Diagram



Component Diagram



Deployment Diagram



Project Time Line

S.No	Functionality	Status (Completed /in-progress/Not started)
01	Abstract Seminar	Completed
02	Design Seminar	
03	Implementation Seminar	
04	Documentation & Paper Submission	

Project Time Line

S.No	Functionality	Status (Completed /in-progress/Not started)
01	Abstract Seminar	Completed
02	Design Seminar	
03	Implementation Seminar	
04	Documentation & Paper Submission	

References

- ❖ **Jarkas, O., Ko, R., Dong, N., Mahmud, R. (2025).** *A Container Security Survey: Exploits, Attacks, and Defenses.* ACM Computing Surveys.
- ❖ **Durieux, T. (2024).** *Empirical Study of the Docker Smells Impact on the Image Size.* IEEE/ACM ICSE.
- ❖ **Thiyagarajan, G., Nayak, P. (2025).** *Docker Under Siege: Securing Containers in the Modern Era.* Study on modern container security risks and practices.
- ❖ Peiyu Liu et al. (2020), “Understanding the Security Risks of Docker Hub”, ESORICS (Springer). This study highlights different risks developers face when using images from public registries.
- ❖ **Nousias, A., et al. (2025).** *Malware Detection in Docker Containers: An Image is Worth a Thousand Logs.* Research article on ML-based detection in container images.
- ❖ Rui Shu, Xiaohui Gu, William Enck (2017), “A Study of Security Vulnerabilities on Docker Hub”, ACM CODASPY. This work scans Docker Hub images and finds many public images with known vulnerabilities.
- ❖ **Zeng, W., Fan, R., Wang, Z., Xiao, Y., Huang, R., Liu, M. (2023).** *Research on Docker Container Network Isolation and Security Management.* ACM CNML.